Memes: the new replicators

***p. 248** I would put my money on one fundamental principle … all life evolves by the differential survival of replicating entities.*

My wager that all life, everywhere in the universe, would turn out to have evolved by Darwinian means has now been spelled out and justified more fully in my paper 'Universal Darwinism' and in the last chapter of *The Blind Watchmaker*. I show that all the alternatives to Darwinism that have ever been suggested are in principle incapable of doing the job of explaining the organized complexity of life. The argument is a general one, not based upon particular facts about life as we know it. As such it has been criticized by scientists pedestrian enough to think that slaving over a hot test tube (or cold muddy boot) is the only method of discovery in science. One critic complained that my argument was 'philosophical', as though that was sufficient condemnation. Philosophical or not, the fact is that neither he nor anybody else has found any flaw in what I said. And 'in principle' arguments such as mine, far from being irrelevant to the real world, can be *more* powerful than arguments based on particular factual research. My reasoning, if it is correct, tells us something important about life everywhere in the universe. Laboratory and field research can tell us only about life as we have sampled it here.

***p. 249** Meme*

The word meme seems to be turning out to be a good meme. It is now quite widely used and in 1988 it joined the official list of words being considered for future editions of Oxford English Dictionaries. This makes me the more anxious to repeat that my designs on human culture were modest almost to vanishing point. My true ambitions—and they are admittedly large—lead in another direction entirely. I want to claim almost limitless power for slightly inaccurate self-replicating entities, once they arise anywhere in the universe. This is because they tend to become the basis for Darwinian selection which, given enough generations, cumulatively builds systems of great complexity. I believe that, given the right conditions, replicators automatically band together to create systems, or machines, that carry them around and work to favour their continued replication. The first ten chapters of *The Selfish Gene* had concentrated exclusively on one kind of replicator, the gene. In discussing memes in the final chapter I was trying to make the case for replicators in general, and to show that genes were not the only members of that important class. Whether the milieu of human culture really does have what it takes to get a form of Darwinism going, I am not sure. But in any case that question is subsidiary to my concern. Chapter 11 will have succeeded if the reader closes the book with the feeling that DNA molecules are not the only entities that might form the basis for Darwinian evolution. My purpose was to cut the gene down to size, rather than to sculpt a grand theory of human culture.

***p. 249** … memes should be regarded as living structures, not just metaphorically but technically.*

DNA is a self-replicating piece of hardware. Each piece has a particular structure, which is different from rival pieces of DNA. If memes in brains are analogous to genes they must be self-replicating brain structures, actual patterns of neuronal wiring-up that reconstitute themselves in one brain after another. I had always felt uneasy spelling this out aloud, because we know far less about brains than about genes, and are therefore necessarily vague about what such a brain structure might actually be. So I was relieved to receive recently a very interesting paper by Juan Delius of the University of Konstanz in Germany. Unlike me, Delius doesn't have to feel apologetic, because he is a distinguished brain scientist whereas I am not a brain scientist at all. I am delighted, therefore, that he is bold enough to ram home the point by actually publishing a